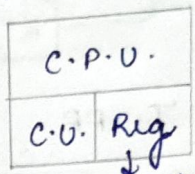


September 21, 23

UNIT-3

Register

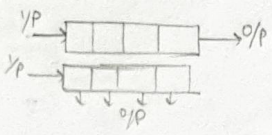
→ It is a group of flip-flops used to store data temporarily during the execution of the program, and to store n-bit data we require n flip-flops.



Types of Registers:

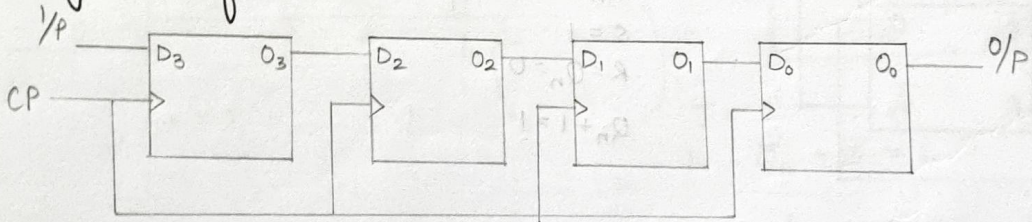
- * S → serial → one bit at a time
- * P → parallel → ~~more~~ multiple bits at a time (simultaneously).

1. SISO
2. SIPO
3. PISO
4. PIPO



* Serial In Serial Out (SISO) Shift register

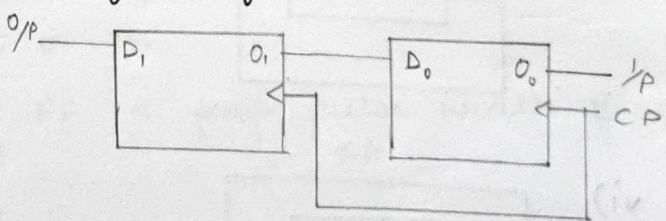
→ Right Shift.



* In exams, take data which is palindrome (1001, 11011) etc.
1011

CP	O ₃	O ₂	O ₁	O ₀
0	0	0	0	0
1	1	0	0	0
2	1	1	0	0
3	0	1	1	0
4	1	0	1	1

Left Shift



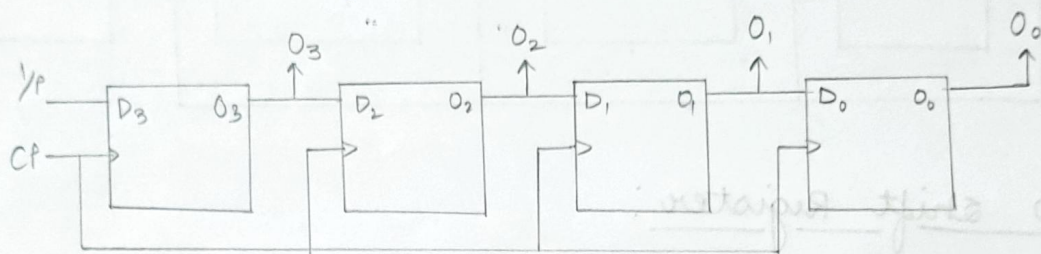
* Time delay = $n * t_{pdf}$

n → no. of flip flop

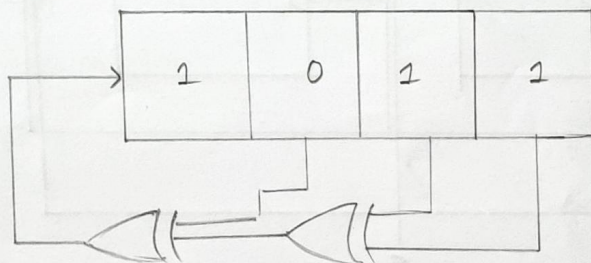
t_{pdf} → time propagation delay of ff.

* SIPO Shift Register:

→ In this data is given serially but output is taken parallelly.



Q. In following SIPO shift register, after how many CP we will get 1011 again?



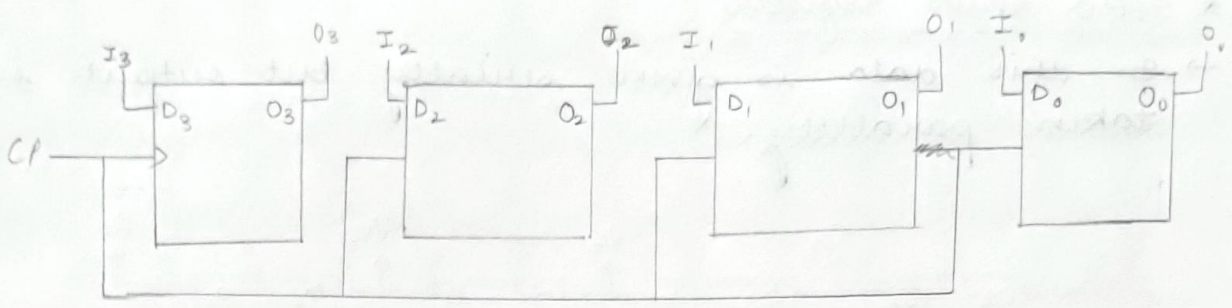
CP	Q ₃	Q ₂	Q ₁	Q ₀
0	1	0	1	1
1	0	1	0	1
2	0	0	1	0
3	1	0	0	1
4	1	1	0	0
5	1	1	1	0
6	0	1	1	1
7	1	0	1	1

* PIPO Shift Register:-

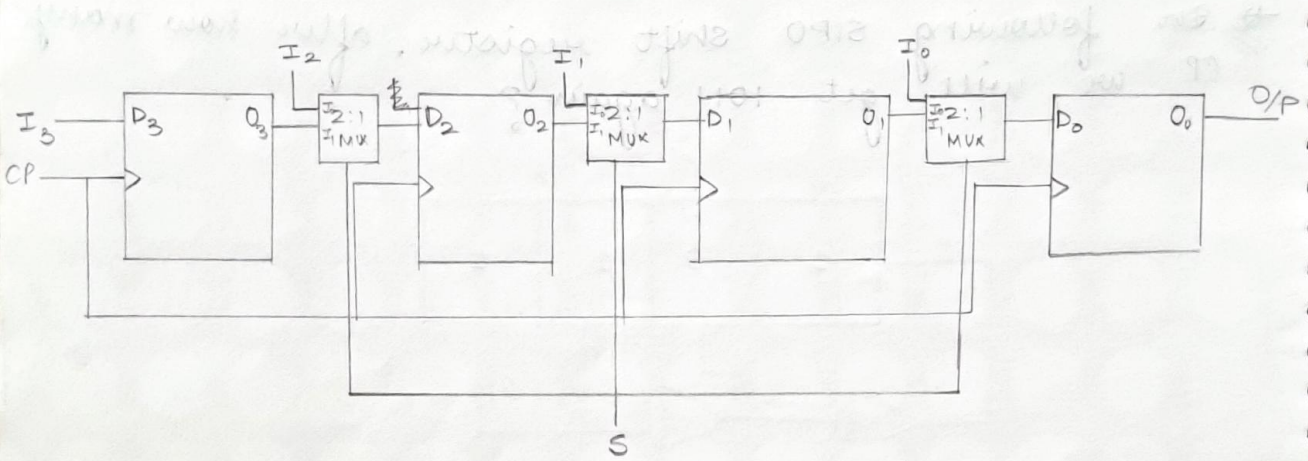
→ Input is given parallel as well as output is taken parallel.

→ fastest.

Time delay = t_{pdf} .



* PISO Shift Register:



→ If $S=0$, I_0 is selected it works as parallel I/O.
 → If $S=1$, I_1 serial I/O.

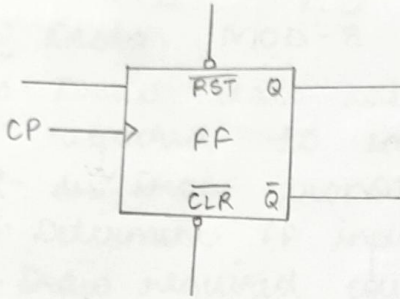
- * Universal Shift Register
 - * Synchronous Counter
- } Youtube.

1	1	0	1	0
1	0	1	0	1
0	1	0	1	0
0	0	0	1	0
0	0	1	1	1
0	1	1	1	1
1	1	1	0	0
1	1	0	1	1

12/10/23

Asynchronous

CP		
-ve	Q	up
+ve	Q	down
-ve	$\bar{Q}$	down
+ve	$\bar{Q}$	up



Output freq:-

$$f_o = \frac{f_i}{N}$$

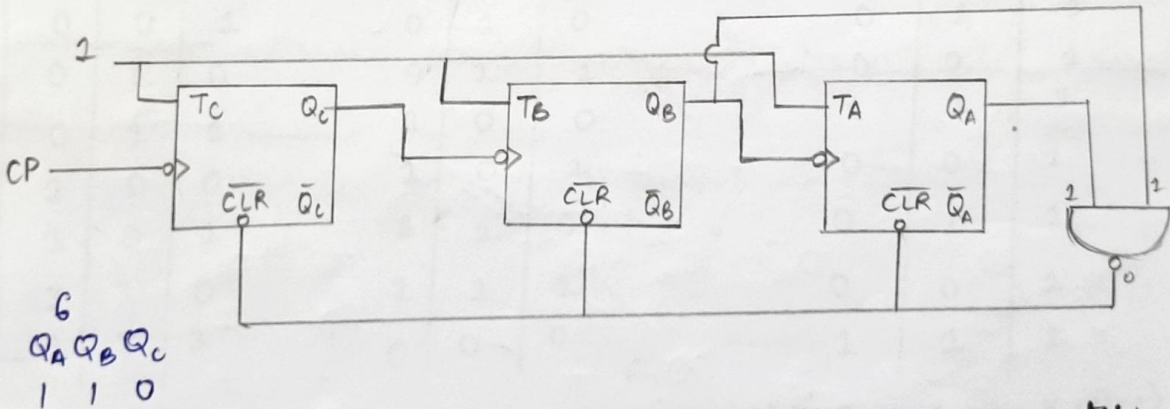
$$100 \times 10^3 \text{ KH} = \frac{2 \times 10^6 \text{ MHZ}}{N}$$

$$N = \text{MOD} - 10$$

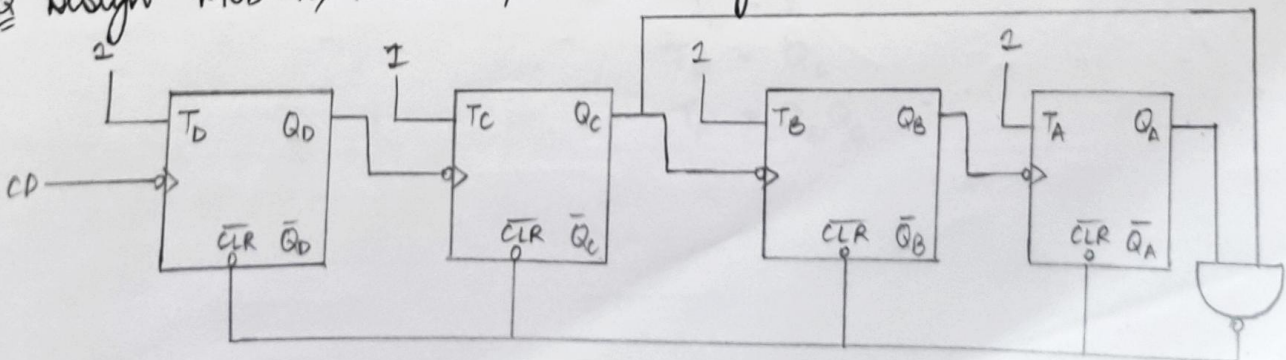
CLR	RST	O/P
0	1	0 clear
1	0	1 set
1	1	FF works normally

Non-binary asynchronous counter:
 $\rightarrow N \neq 2^n$

Q Design MOD-6 asynchronous counter / ripple / serial.
 $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 0$ (Note: 6 takes a short period of time due to pd)



Q Design MOD-10 / decimal / decade asynchronous counter.



18/20/23

MOD-N $\rightarrow$ N = No. of CP applied

Synchronous or Parallel Counter

- $\rightarrow$ In this common CP is applied to all the flip-flops simultaneously.
- $\rightarrow$ In this we can design any pattern of count sequence

Q Design MOD-8 synchronous counter.

1. Draw state table & determine no. of flip flops required to implement that counter.
2. Draw transition table.
3. Determine FF inputs by the help of K-map.
4. Draw required counter with the help of FF and suitable logic gates.

MOD-8

0-7

$8 \leq 2^3$

Present state

Q_A	Q_B	Q_C
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

Next state

Q_A^+	Q_B^+	Q_C^+
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1
0	0	0

$Q_A \oplus Q_A^+$

T_A	T_B	T_C
0	0	1
0	1	1
0	0	1
1	1	1
0	0	1
0	1	1
0	0	1
1	1	1

On solving, (with K-Map)

$$T_C = 1$$

$$T_B = Q_C$$

$$T_A = Q_B \cdot Q_C$$

Q_A	Q_B	Q_C	T_A	T_B	T_C
0	0	0	d	d	d
0	0	1	0	1	1
0	1	0	d	d	d
0	1	1	1	0	1
1	0	0	d	d	d
1	0	1	1	1	1
1	1	0	d	d	d
1	1	1	0	0	1

Q_A^+ Q_B^+ Q_C^+
 D_A D_B D_C

→ LOCK-OUT CONDITION

→ When counter goes from used to unused state and unused state to another unused state, then it will never go to used state.

$0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6 \rightarrow 7 \rightarrow 0$
 $3 \rightarrow 4 \rightarrow 5 \rightarrow 6 \rightarrow 7 \rightarrow 0$ → unused state.

This is known as lock out condition.

→ To avoid lock-out condition, write down unused state and their next state will be used state.

19 October, 23

Ring Counter:

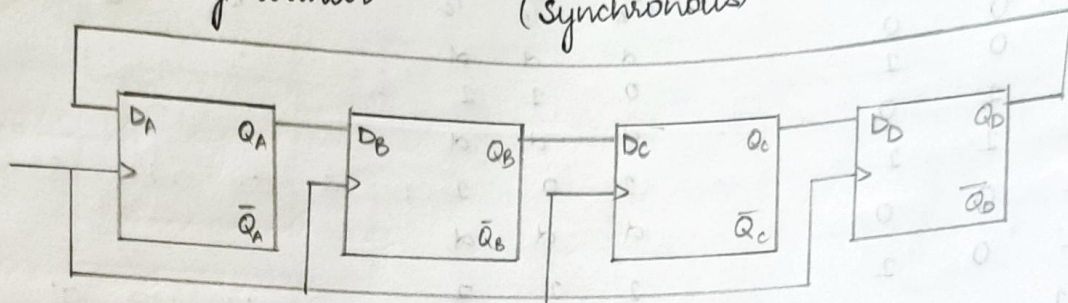
- In this uncomplemented output of last flip-flop is connected to input of first flip flop.
- In this any one flip-flop output is high at a time.

* n-bit

n → no. of flip flops.

→ 4 bit ring counter

(Synchronous)



→ Any one output high.

8 → 4 → 2 → 1 → 8

CP	QA	QB	QC	QD
0	1	0	0	0
1	0	1	0	0
2	0	0	1	0
3	0	0	0	1
4	1	0	0	0

* Total used states → $n = \text{MOD}-N$

* Total unused states → $2^n - n$
 $2^4 - 4 = 12$

$$* f_o = \frac{f_i}{N}$$

In J-K Flipflop,

$$J = S = D$$

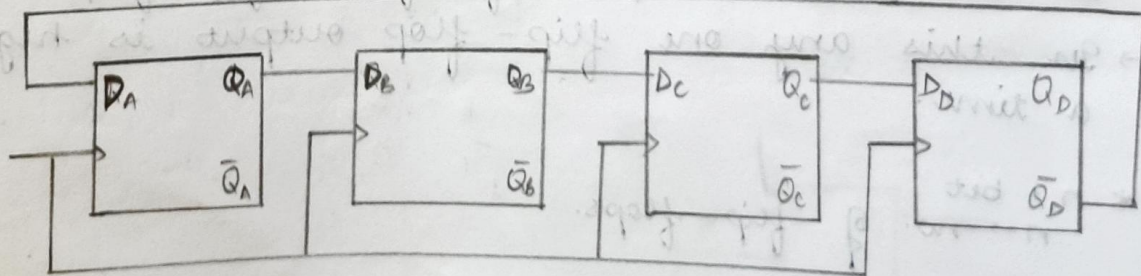
$$K = R = \bar{D}$$

switch Tail Counter/Mobile Counter.

Johnson Counter : / Creeping Counter / Twisted Ring Counter

→ In this complemented output of last flip-flop is connected to last flip input of first flip-flop.

* 4-bit (n-bit) Johnson counter:



CP	Q_A	Q_B	Q_C	Q_D
0	0	0	0	0
1	1	0	0	0
2	1	1	0	0
3	1	1	1	0
4	1	1	1	1
5	0	1	1	1
6	0	0	1	1
7	0	0	0	1
8	0	0	0	0

* Total used states: $2^n = \text{MOD } 8$

* Total unused states: $2^n - 2n$

$$= 2^4 - 8$$

$$= 8$$

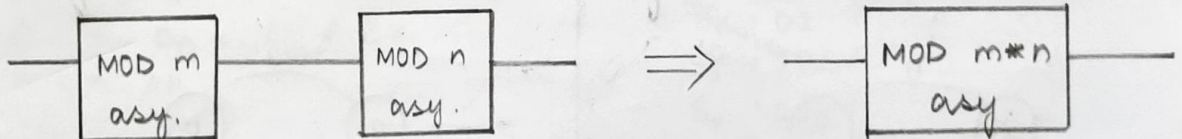
$$* f_o = \frac{f_i}{N} = \frac{f_i}{8}$$

Q If input frequency is f_i , what will be f_o and total no. of unused states in 5-bit Johnson counter.

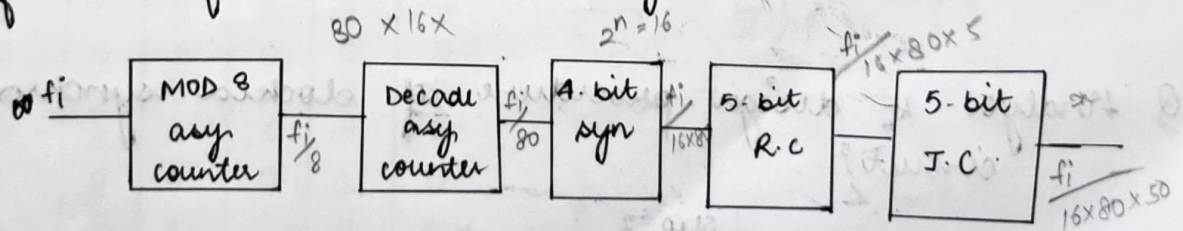
$$\rightarrow f_o = \frac{f_i}{10}$$

$$\text{unused states} = 32 - 10 = 22$$

Q If f_o of creeping counter is 100 kHz & $f_i = 1 \text{ MHz}$ Design creeping counter.



Q If $f_i =$, find f_o in following circuit?

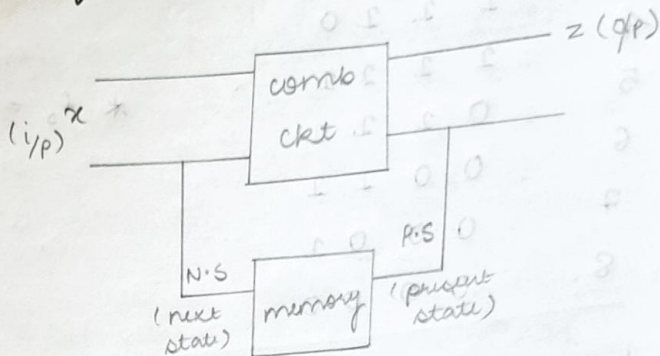


$$\rightarrow 64000$$

21 October, 23

clocked synchronous sequential ckt.

→ It is described by Moore and Mealy machine.



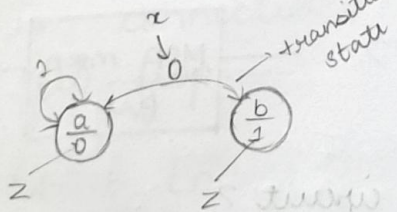
Moore Model

1. Output z depends upon present states only. It doesn't depend upon input x .

$$z = f(Q)$$

2. If input changes, output doesn't change.

3. More states are required

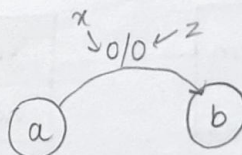


Mealy model

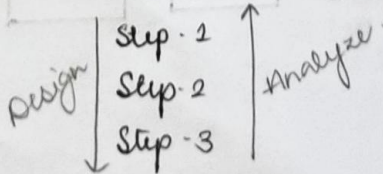
1. Output z depends upon present state as well as input x .

$$z = f(Q, x)$$

2. Output changes with the input.
3. Less states are required.



Q Analyse & design procedure of clocked synchronous circuit.



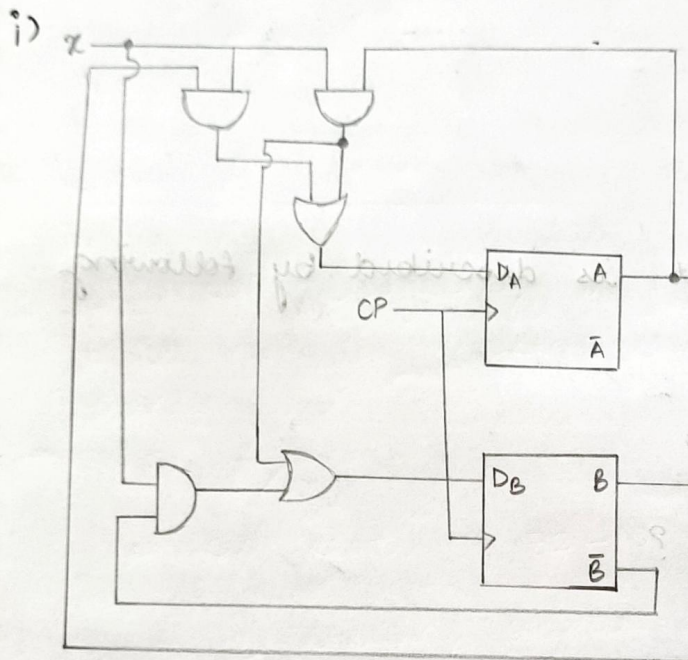
Q A clocked synchronous sequential ckt is described by following equations:

$$D_A = Ax + Bx$$

$$D_B = Ax + \bar{B}x$$

$$Z = AB$$

- Draw the logical diagram.
- Draw state diagram & state table.
- state diagram?



$Z = AB$
 → Output doesn't depend upon x .
 Hence, it is an example of moore machine.

logical diagram

ii)

$$A^+ = D_A = Ax + Bx$$

$$B^+ = D_B = Ax + \bar{B}x$$

$$Z = AB$$

iii)

$$S_0 \rightarrow 00$$

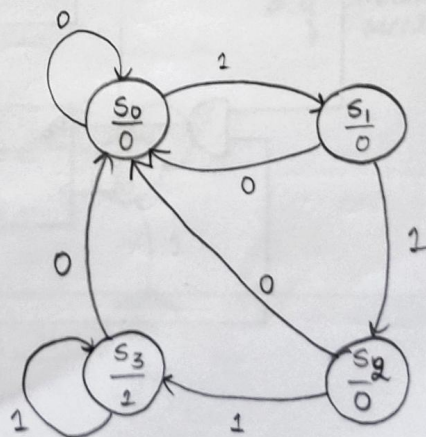
$$S_1 \rightarrow 01$$

$$S_2 \rightarrow 10$$

$$S_3 \rightarrow 11$$

i/p x	P.S		N.S		O/P Z
	A	B	A ⁺	B ⁺	
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	0	0	0
0	1	1	0	0	1
1	0	0	0	1	0
1	0	1	1	0	0
1	1	0	1	1	0
1	1	1	1	1	1

State table



State diagram

Q Design a clocked synchronous sequential ckt which detects three or more consecutive ones. (2 D-flip flops)

- Draw state diagram.
- state table.
- logical diagram of clocked seq. syn. ckt.

$$D_A = A^+ = \sum m(5, 6, 7)$$

$$D_B = B^+ = \sum m(4, 6, 7)$$

$$Z = \sum m(3, 7)$$

using K-map,

$$Z = AB$$

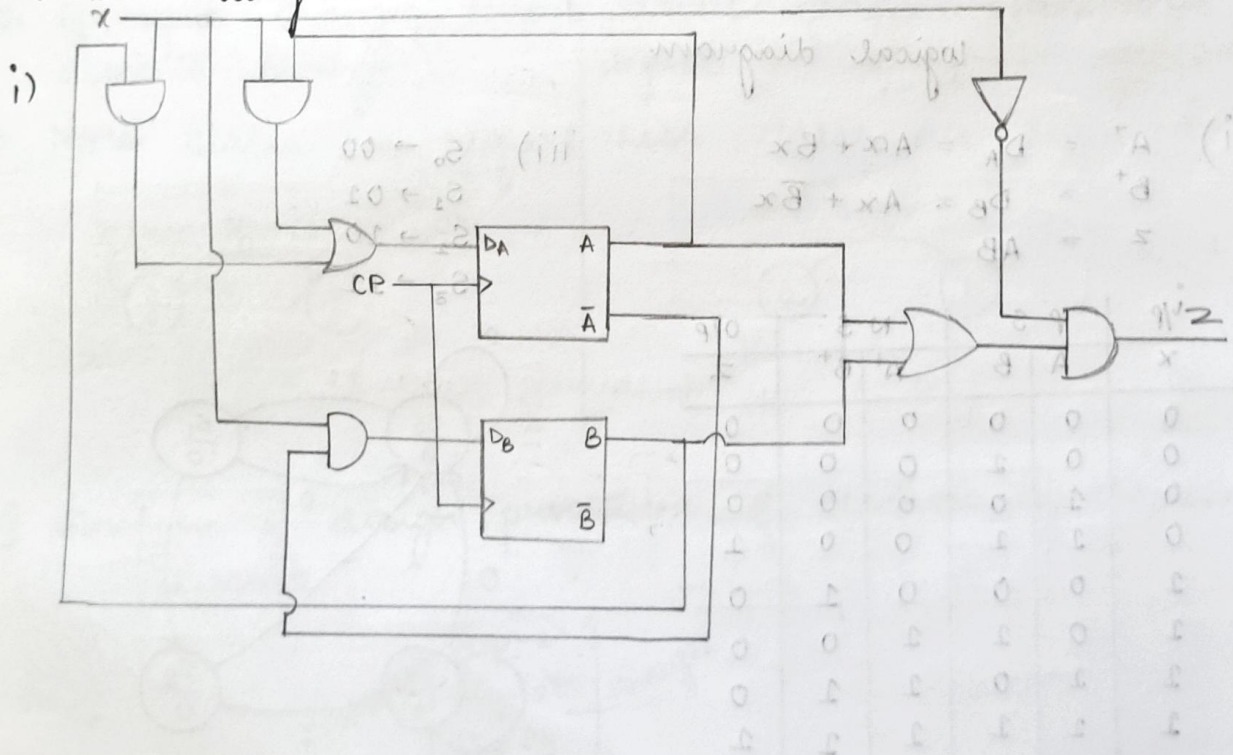
Q A clocked syn. seq. ckt. is described by following eqⁿ,

$$D_A = Ax + Bx$$

$$D_B = \bar{A}x$$

$$Z = (A+B)\bar{x}$$

- Draw logical diagram.
- ~~state~~ Milroy or Moore?
- state diag table
- state diagram.



ii) output depends upon input. Mileay machine.

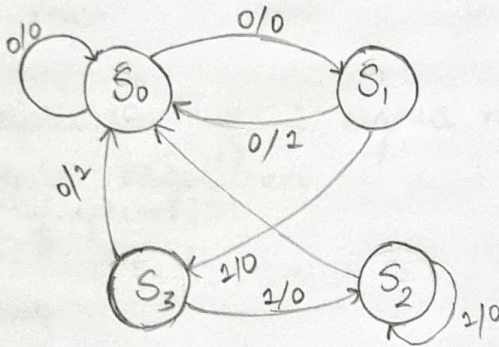
iii)

x	A	B	A ⁺	B ⁺	z
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	0	1
0	1	1	0	0	1
1	0	0	0	1	0
1	0	1	1	1	0
1	1	0	1	0	0
1	1	1	1	0	0

$$A^+ = D_A$$

$$B^+ = D_B$$

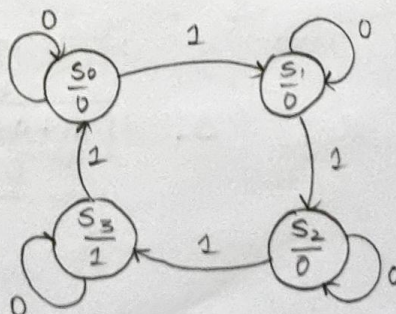
iv)



Q A clocked syn seq ckt has one control line (input) when its value is 1, it goes to next state. (works as 2 bit up counter) otherwise it remains in same state. using D FF, draw state diagram, state table & logical diagram. when it has max value, outcome becomes high otherwise low.

→ A = S₀ = 00
 B = S₁ = 01
 C = S₂ = 10
 D = S₃ = 11

Moskey
 if reaches
 Mealy.



State Reduction Table.

→ From going present state to next state, if next states and outputs are same, both states are called equivalent states and we can eliminate any one state and corresponding table is known as reduced state table & corresponding state diagram is called reduced state diagram.

P.S.	N.S.		O/P	
	x=0	x=1	x=0	x=1
a	d	c	1	0
b	c	d	0	1
c	c	a	1	0
d	c	c	1	0
e	d	c	1	0